

The empirical recurrence for OEIS A229605, recovered from its tail

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Abstract

OEIS A229605 records “Empirical recurrence of order 46” and keeps the recurrence itself in a linked file rather than in the entry, so the conjecture cannot be read off the entry at all. It can still be settled. The entry counts arrays under a condition local to a bounded window of lines, which makes the count a walk count on a finite digraph and forces a monic integer recurrence of order at most the number $S' = 599$ of states with distinct futures. Berlekamp–Massey on enough exact terms therefore returns the minimal such recurrence, and on the tail beginning at $n = 2$, so the recurrence is proved for every $n \geq 48$. Its last coefficient is $c_{46} = -4194304 \neq 0$, so no further tail satisfies anything shorter; any order-46 recurrence the sequence satisfies from any point on must then have the same characteristic polynomial. The entry’s recurrence is the one displayed here, whatever its linked file contains, and it is proved.

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1 A conjecture that is not written down

OEIS A229605 is “Number of defective 3-colorings of an $n \times 70..2$ array connected horizontally and vertically with exactly two mistakes, and colors introduced in row-major 0..2 order.”. It has offset 1 and begins

120, 15795, 792929, 31848813, 1153135492, 39229935270, 1279020266434, 40413033646242, ...

The entry states:

Empirical recurrence of order 46 (see link above).

As of the “Last modified” line on the live entry (Apr 27 2021 21:29:04, revision 9) this is still recorded as empirical, and it is still open. The coefficients are nowhere in the entry text: what is conjectured is only that a recurrence of that order exists.

2 The count is a walk count

The entry’s condition is local. It constrains a bounded window of consecutive lines of the array and nothing beyond it, so the admissible configurations of one window are the vertices of a finite digraph, an edge joins u to v when v may follow u , and an admissible array is exactly a walk. Writing M for the adjacency matrix and ι, τ for the indicator vectors of the admissible first and last windows,

$$a(n) = \iota^\top M^{j(n)} \tau$$

for a linear reindexing j fixed by the entry’s shape. States with identical futures contribute identically to that product and may be merged without changing any count; after merging $S' = 599$ states remain, and it is that bound every computation below uses.

Lemma 1. *a satisfies a monic linear recurrence with integer coefficients of order at most $S' = 599$.*

Proof. By the Cayley–Hamilton theorem $M^{S'}$ is an integer combination of $I, M, \dots, M^{S'-1}$, and multiplying that relation on the left by $\iota^\top M^j$ and on the right by τ turns it into a relation among $S' + 1$ consecutive values of a . $\square$

3 Why the question has to be asked of a tail

Running Berlekamp–Massey on the sequence from its first term returns an order LARGER than 46: the opening terms do not satisfy the recurrence, and a recurrence that holds only eventually always looks longer when the exceptional terms are included. That is not evidence against the entry. An OEIS empirical recurrence is asserted for n beyond some index, not from the first term, so the question has to be asked of the tails. It was asked of each tail in turn, and the tail beginning at $n = 2$ is the first whose minimal order is exactly the stated 46.

Lemma 2. *A sequence known to satisfy some linear recurrence of order at most S' is determined, together with its minimal recurrence, by any $2S'$ consecutive terms: Berlekamp–Massey applied to them returns that minimal recurrence exactly.*

Applied to $2S' = 1198$ exact terms from $n = 2$ on, it returns order 46 — the order the entry states — with integer coefficients:

$$a(n) = \sum_{i=1}^{46} c_i a(n-i),$$

c_1	160	c_{13}	−4033767350861204	c_{25}	25527561834086817792	c_{37}	36312418181423104
c_2	−11568	c_{14}	25297498699954024	c_{26}	−27102934386893912576	c_{38}	−9723805472423936
c_3	502856	c_{15}	−117862915123711792	c_{27}	18960827663950593472	c_{39}	1997753026478080
c_4	−14753692	c_{16}	434786763413873360	c_{28}	−6796440560751387904	c_{40}	−317217815592960
c_5	310691338	c_{17}	−1300948573049680764	c_{29}	−2429531414338093888	c_{41}	38696465989632
c_6	−4865580145	c_{18}	3180279929905216672	c_{30}	5670349522978748672	c_{42}	−3562472996864
c_7	57706571017	c_{19}	−6319604334706054696	c_{31}	−4502788747653985152	c_{43}	239515729920
c_8	−518811804838	c_{20}	9994434537835615296	c_{32}	2109183503909061120	c_{44}	−11097079808
c_9	3437274129173	c_{21}	−11886111892798187360	c_{33}	−467739843639346688	c_{45}	316669952
c_{10}	−14922597484786	c_{22}	8748561682530863200	c_{34}	−139357589983404032	c_{46}	−4194304
c_{11}	16023166271135	c_{23}	1048853866818654320	c_{35}	188967994964873216		
c_{12}	379233089298668	c_{24}	−14749562939522977728	c_{36}	−100875649184321536		

4 The recurrence, and the identification

Theorem 3. *The recurrence above holds for every $n \geq 48$, and it is the recurrence OEIS A229605 records.*

Proof. That it holds is Lemma 2 applied to the tail from $n = 2$, whose minimal recurrence it is.

For the identification, write $q_{\min}$ for the characteristic polynomial of that minimal recurrence, $x^{46} - c_1 x^{45} - \dots - c_{46}$. Its constant term is $-c_{46} = 4194304$, which is not zero, so $q_{\min}(0) \neq 0$ and $q_{\min}$ has no root at the origin. A solution of a constant-coefficient recurrence is a combination of terms $n^k \lambda^n$ with λ a root of its characteristic polynomial, and only the components with $\lambda = 0$ vanish after finitely many terms. Since $q_{\min}$ has none, no component of a dies out, and the minimal recurrence of every further tail is again $q_{\min}$ — the order cannot drop below 46 at any later starting point.

Now let q be the characteristic polynomial of whatever recurrence the entry asserts. The entry asserts it has order 46, so $\deg q = 46$, and it is monic. It holds from some index t on. From $\max(t, 2)$ on, the sequence satisfies both q and $q_{\min}$, and the minimal polynomial of that tail is $q_{\min}$ by the previous paragraph; therefore $q_{\min} \mid q$. Both are monic of degree 46, so $q = q_{\min}$. $\square$

5 Verification

Four checks, all in exact integer or rational arithmetic.

First, the walk model reproduces all 14 terms the entry publishes, exactly. This is what ties the model to the entry, and it is the only place where reading the entry’s English enters the argument.

Second, the order was computed twice by independent routes: once modulo a 61-bit prime, where every intermediate is a machine word, and once over $\mathbb{Q}$ in exact rational arithmetic. Both return 46, and the exact run additionally confirms every coefficient is an integer — had any been a proper fraction the recurrence would not be the entry’s.

Third, the recovered recurrence was evaluated directly on the entry’s own published terms, with no matrices and no Berlekamp–Massey involved. It holds at each of the 0 indices where the entry publishes enough earlier terms to test it.

Fourth, the last coefficient was checked to be nonzero, which is what the identification above turns on; it is -4194304 .

Remark 4. The conjecture’s own statement was never read. The entry pins it down to the family of order-46 recurrences this sequence satisfies from some point on, and that family turns out to have exactly one member.

References

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